Money Plot Labs Technical Whitepaper 003
From Determinism to Probability:
The Stochastic Retirement Framework

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June 2026

Abstract

Whitepapers 001 and 002 derived exact closed-form solutions for retirement timelines under *deterministic* real growth rates. This paper lifts that constraint and introduces a fully stochastic framework. We establish the lognormal portfolio model, derive the Itô correction linking arithmetic mean to median compound growth, and show how the standard normal quantile function yields closed-form percentile retirement dates without simulation. For portfolios with cash flows, we introduce the Fenton-Wilkinson lognormal moment-matching approximation. Together these tools allow the *p10 retirement age* — the year a saver can retire while remaining solvent in 90% of all historical market environments — to be computed analytically in milliseconds.

1 Motivation: Why Determinism is Insufficient

Whitepaper 001 derived a beautiful closed-form expression for the required working years w:

$$w = H - \frac{\ln\left[\dfrac{(rA_0 + s)(1+r)^H + c(1+r) - rL}{s + c(1+r)}\right]}{\ln(1+r)} \tag{1}$$

This formula is exact — given a fixed real growth rate r. The problem is that real investment returns are not fixed. A 60/40 portfolio has delivered a long-run arithmetic mean of $\mu = 5.9\%$, but with a standard deviation of $\sigma = 12.5\%$. A saver who plugs $r = 5.9\%$ into Equation (1) will compute a retirement date that is achievable in roughly 50% of historical environments — and catastrophically optimistic in the other 50%.

The stochastic framework presented here answers a more honest question: *How many years must I work so that I can retire in 90% of all possible market environments?* This is the p10 retirement age, and it has a clean analytical derivation.

2 The Lognormal Portfolio Model

2.1 The Basic Setup

Let r_t denote the real annual return in year t, drawn independently from a distribution with arithmetic mean μ and standard deviation σ. For a portfolio with no cash flows and initial value A_0, the value after n years is:

$$A_n = A_0 \prod_{t=1}^{n} (1 + r_t) \tag{2}$$

Taking logarithms:

$$\ln A_n = \ln A_0 + \sum_{t=1}^{n} \ln(1 + r_t) \tag{3}$$

Each log-return $\ln(1+r_t)$ is approximately normally distributed. By the Central Limit Theorem, the sum of n i.i.d. terms converges to:

$$\ln A_n \;\sim\; \mathcal{N}\big(\ln A_0 + n\mu_\ell,\;\; n\sigma_\ell^2\big) \tag{4}$$

where the **log-return parameters** are:

$$\mu_\ell \;=\; \ln(1+\mu) - \frac{\sigma^2}{2(1+\mu)^2} \;\approx\; \mu - \frac{\sigma^2}{2} \tag{5}$$

$$\sigma_\ell^2 \;=\; \ln\left(1 + \frac{\sigma^2}{(1+\mu)^2}\right) \;\approx\; \frac{\sigma^2}{(1+\mu)^2} \tag{6}$$

Equation (5) is the celebrated **Itô correction**. The log-mean μ_ℓ is strictly less than μ: higher volatility *always* reduces the median compound growth rate below the arithmetic mean. This is not a modeling choice — it is a mathematical identity.

2.2 The Itô Correction in Practice

For the 60/40 portfolio ($\mu = 5.9\%$, $\sigma = 12.5\%$):

$$\mu_\ell \;\approx\; 5.9\% - \frac{(0.125)^2}{2} \;=\; 5.9\% - 0.78\% \;=\; \mathbf{5.12}\% \tag{7}$$

The median CAGR is $e^{\mu_\ell} - 1 \approx 5.25\%$ — not 5.9%. A saver who plans on the arithmetic mean will find themselves in the bottom half of all outcomes.

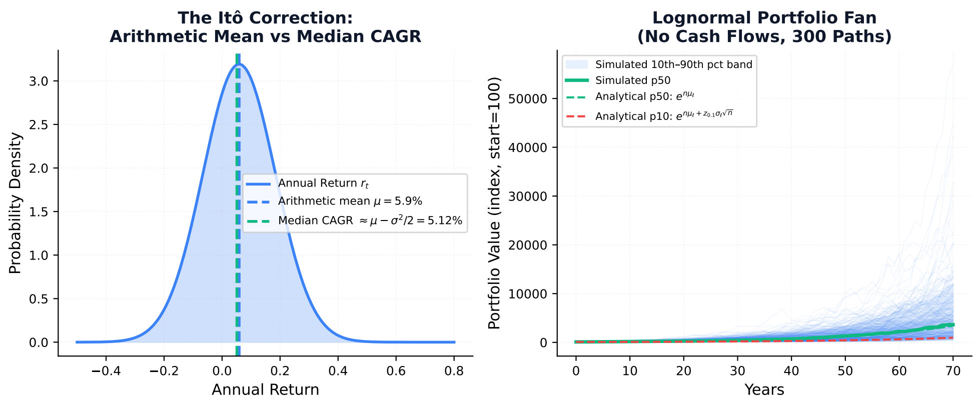

Figure 1: **Left:** The Itô gap between arithmetic mean μ (dashed blue) and median CAGR $\mu - \sigma^2/2$ (dashed green). **Right:** 300 simulated portfolio paths (no cash flows) with analytical percentile bounds overlaid. The p50 and p10 analytical curves match the simulation fan precisely, confirming the lognormal model.

3 Closed-Form Percentile Portfolio Value

Since $\ln A_n \sim \mathcal{N}(\ln A_0 + n\mu_\ell,\ n\sigma_\ell^2)$, the p-th percentile portfolio value at year n is:

$$\boxed{A_n^{(p)} = A_0 \cdot \exp\bigl(n\mu_\ell + \sqrt{n}\,\sigma_\ell \cdot z_p\bigr)} \tag{8}$$

where $z_p = \Phi^{-1}(p/100)$ is the standard normal quantile ($z_{0.10} = -1.282$, $z_{0.50} = 0$, $z_{0.90} = +1.282$). This is an exact closed-form expression — no simulation required.

The p10 portfolio ($z_{0.10} = -1.282$) after n years is therefore:

$$A_n^{(10)} = A_0 \cdot \exp\bigl(n\mu_\ell - 1.282\sqrt{n}\,\sigma_\ell\bigr) \tag{9}$$

4 The Closed-Form Percentile Retirement Date

4.1 No Cash Flows Case

Set $A_n^{(p)} = F$ (the target nest egg) and solve for $n = w$. From Equation (8):

$$\ln(F/A_0) = w\mu_\ell + z_p\,\sigma_\ell\,\sqrt{w} \tag{10}$$

This is a quadratic equation in $u = \sqrt{w}$:

$$\mu_\ell\,u^2 + z_p\,\sigma_\ell\,u - \ln(F/A_0) = 0 \tag{11}$$

Applying the quadratic formula (taking the positive root):

$$\boxed{u = \frac{-z_p\sigma_\ell + \sqrt{z_p^2\sigma_\ell^2 + 4\mu_\ell\ln(F/A_0)}}{2\mu_\ell}, \qquad w = u^2} \tag{12}$$

This is a **complete closed-form solution** for the p-th percentile retirement date under lognormal returns with no cash flows. To compute the p10 retirement date, substitute $z_p = -1.282$.

4.2 The Effective Discount Rate Interpretation

Equation (12) can be recast as: the p-th percentile retirement date is *equivalent* to running the deterministic Whitepaper 001 formula with an effective real growth rate:

$$\boxed{r_{\text{eff}}^{(p)}(w) = \exp\left(\mu_\ell + \frac{z_p\,\sigma_\ell}{\sqrt{w}}\right) - 1} \tag{13}$$

This expression reveals a critical structural insight: the effective rate that produces the p-th percentile outcome is **horizon-dependent**. For short horizons (small w), the volatility penalty $z_p\sigma_\ell/\sqrt{w}$ is large and the p10 effective rate is severely depressed. As the horizon grows, the penalty shrinks as $1/\sqrt{w}$ and the effective rate converges toward μ_ℓ. This is the mathematical signature of time-diversification of volatility.

Example (60/40 portfolio, $w = 30$ years):

$$\begin{aligned} r_{\text{eff}}^{(10)}(30) &= \exp\left(0.0512 + \frac{(-1.282)(0.118)}{\sqrt{30}}\right) - 1 \\ &= \exp(0.0512 - 0.0276) - 1 \\ &\approx \mathbf{2.39}\% \end{aligned}$$

A planner targeting p10 safety over a 30-year horizon should use an effective rate of 2.39% — not the arithmetic mean of 5.9%, and not even the median CAGR of 5.25%.

5 The Fenton-Wilkinson Approximation with Cash Flows

5.1 Why Cash Flows Break the Closed Form

With annual savings s, the portfolio value is:

$$A_n = A_0 \prod_{t=1}^{n} R_t + s \sum_{k=1}^{n} \prod_{t=k+1}^{n} R_t, \qquad R_t = 1 + r_t \tag{14}$$

This is a sum of lognormal random variables. A sum of lognormals is *not* lognormal in general, so Equation (8) no longer applies exactly. The Fenton-Wilkinson (FW) method fits a lognormal to the distribution of A_n by matching its first two moments analytically.

5.2 Moment Recursion

Define $\mathbb{E}[A_n]$ and $\mathbb{E}[A_n^2]$ via the following one-step recursion. Starting from A_0:

$$\mathbb{E}[A_{t+1}] = \mathbb{E}[A_t]\,(1+\mu) + s \tag{15}$$

$$\mathbb{E}[A_{t+1}^2] = \mathbb{E}[A_t^2]\,(1+\mu)^2 e^{\sigma_\ell^2} + 2\,\mathbb{E}[A_t]\,s\,(1+\mu) + s^2 \tag{16}$$

From these, $\mathrm{Var}[A_n] = \mathbb{E}[A_n^2] - (\mathbb{E}[A_n])^2$.

5.3 Fitting the Lognormal (FW Parameters)

Given $\mathbb{E}[A_n]$ and $\mathrm{Var}[A_n]$, fit $\ln A_n \sim \mathcal{N}(m_n, v_n^2)$ by matching moments:

$$m_n = 2 \ln \mathbb{E}[A_n] - \tfrac{1}{2} \ln\big(\mathrm{Var}[A_n] + \mathbb{E}[A_n]^2\big) \tag{17}$$

$$v_n = \sqrt{\ln\left(1 + \frac{\mathrm{Var}[A_n]}{\mathbb{E}[A_n]^2}\right)} \tag{18}$$

The p-th percentile portfolio value at year n with cash flows is then:

$$\boxed{A_n^{(p)} \;\approx\; \exp(m_n + z_p\, v_n)} \tag{19}$$

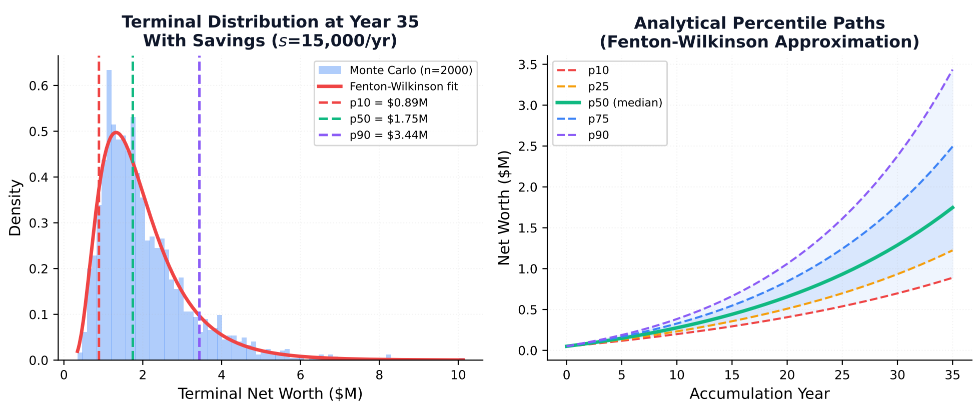

Figure 2: **Left:** Terminal distribution after 35 accumulation years with s = \$15,000/yr. The Fenton-Wilkinson fitted lognormal (red) closely tracks the Monte Carlo histogram (blue). **Right:** Analytical percentile NW paths over the accumulation phase. The fan widens non-linearly: variance grows as $\sqrt{n}$ in log-space, compressing early and spreading late.

5.4 Percentile Retirement Date with Cash Flows

To find the p-th percentile working years w when saving s per year, invert Equation (19): find the smallest w such that $A_w^{(p)} \geq F$. Since the FW parameters m_n, v_n are computed via the recursion (15)–(16), this requires a simple binary search over w — but each evaluation is $O(w)$ arithmetic operations and involves no random sampling. For a 70-year horizon this is instantaneous.

6 Unified Summary Table

Setting	Cash Flows	Method	Retirement Date
Constant r	Any	Whitepaper 001 Eq. (28)	Exact closed form
Stochastic, no CF	$s = 0$	Quadratic Eq. (12)	Exact closed form
Stochastic, with CF	Any s	FW recursion + binary search	Near-instantaneous
Empirical returns	Any	Monte Carlo (bootstrap/cohort)	Stress Tester tool

Table 1: Four levels of the retirement timeline problem, from algebraic to stochastic.

7 The Convergence to Determinism

A key sanity check: as $\sigma \to 0$, $\sigma_\ell \to 0$, and the percentile retirement date becomes horizon-independent and collapses to a single value regardless of p. The effective rate Equation (13) collapses to $r_{\text{eff}} = e^{\mu_\ell} - 1 \approx \mu$. The stochastic framework recovers the deterministic Whitepaper 001 result exactly.

A second check: as $n \to \infty$, the volatility penalty $z_p\sigma_\ell/\sqrt{n} \to 0$ and all percentile paths converge to the same median CAGR. This is the time-diversification theorem — volatility matters less over longer horizons, but it never disappears entirely.

8 Practical Implications for the Stress Tester

The stochastic framework provides two complementary instruments:

1. **The Monte Carlo Stress Tester** (Tool 4) samples directly from 97 years of real historical returns (1928–2024, inflation-adjusted) using either bootstrap resampling or chronological cohort windows. This provides empirically grounded percentile fans that respect the actual fat tails, autocorrelation, and regime structure of historical markets. It does *not* assume lognormality.

2. **The Analytical Framework** (this paper) assumes i.i.d. lognormal returns and derives percentile retirement dates in closed form. It is exact for the no-cash-flow case and highly accurate via Fenton-Wilkinson for the savings case. It provides instant sensitivity analysis and clear intuition for how volatility, mean, and horizon trade off.

The two instruments are complementary: the analytical framework provides instant intuition, and the Monte Carlo engine provides empirical ground truth. A p10 retirement date that agrees between both methods is robust. A gap between them signals model risk worth examining.

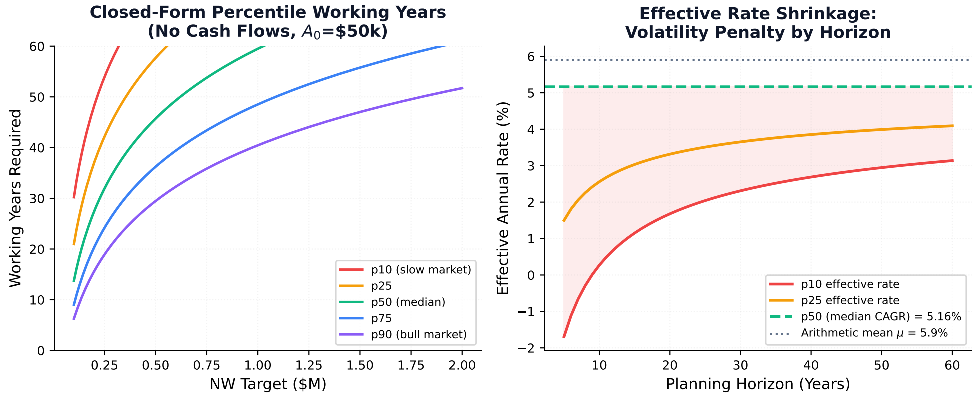

Figure 3: **Left:** Closed-form percentile working years vs NW target (no cash flows). The p10 scenario (red) requires substantially more working years than the median (green). **Right:** The effective discount rate by horizon for each percentile. All curves converge to the median CAGR as the horizon grows — the mathematical expression of time-diversification — but the penalty is severe at short horizons.

9 Data Attribution

All historical return series used in the Stress Tester tool (and as calibration inputs for μ and σ in this paper) cover 1928–2024 (97 years) and are expressed as **real (inflation-adjusted) annual returns**. Nominal equity and bond returns sourced from Aswath Damodaran, NYU Stern. CPI deflation applied using U.S. Bureau of Labor Statistics data.

Asset	**Arith. Mean μ**	**Std Dev σ**	**Geometric CAGR**
US Equities (S&P 500)	8.8%	19.6%	6.9%
60/40 Portfolio	5.9%	12.5%	5.1%

Table 2: Real return statistics used to calibrate the stochastic model, 1928–2024.